glands. Their secretions are called **hormones** which are produced in **minute quantities**. The secretions diffuse into the blood stream and are carried to the distant parts of the body. They act on specific organs which are referred as **target organs**.



The branch of biology which deals with the study of the endocrine glands and its physiology is known as '**Endocrinology**'. **Thomas Addison** is known as Father of Endocrinology. English physiologists **W. M. Bayliss** and **E. H. Starling** introduced the term ***hormone*** in 1909. They first discovered the hormone secretin.

Exocrine glands have specific ducts to carry their secretions e.g. salivary glands, mammary glands, sweat glands.

Endocrine glands present in human and other vertebrates are

a. Pituitary gland
b. Thyroid gland
c. Parathyroid gland
d. Pancreas (Islets of Langerhans)
e. Adrenal gland (Adrenal cortex and Adrenal medulla)
f. Gonads(Testes and Ovary)
g. Thymus gland

16.2.1 Pituitary Gland

The **pituitary gland** or **hypophysis** is a pea shaped compact mass of cells located at the base of the midbrain attached to the hypothalamus by a **pituitary stalk**. The pituitary gland is anatomically composed of two lobes and perform different functions. They are the anterior lobe (**adenohypophysis**) and the posterior lobe (**neurohypophysis**). The intermediate lobe is non-existent in humans.

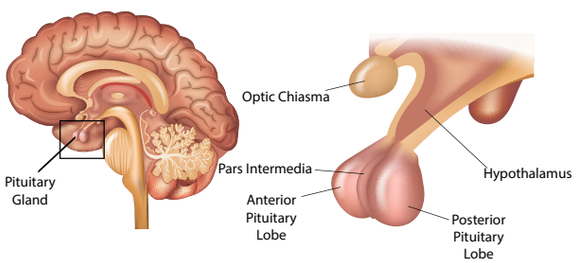


Figure 16.7 Pituitary Gland

The pituitary gland forms the major endocrine gland in most vertebrates. It regulates and controls other endocrine glands and so is called as the "**Master gland**".

Hormones secreted by the anterior lobe (Adenohypophysis) of pituitary

The anterior pituitary is composed of different types of cells and secrete hormones which stimulates the production of hormones by other endocrine glands. The hormones secreted by anterior pituitary are

a. Growth Hormone
b. Thyroid stimulating Hormone
c. Adrenocorticotropic Hormone
d. Gonadotropic Hormone which comprises the Follicle Stimulating Hormone and Luteinizing Hormone
e. Prolactin

a. Growth hormone (GH)

GH promotes the development and enlargement of all tissues of the body. It stimulates the growth of muscles, cartilage and long bones. It controls the cell metabolism.

The improper secretion of this hormone leads to the following conditions.

Dwarfism: It is caused by decreased secretion of growth hormone in children. The characteristic features are stunted growth, delayed skeletal formation and mental disability.

Gigantism: Oversecretion of growth hormone leads to gigantism in children. It is characterised by overgrowth of all body tissues and organs. Individuals attain abnormal increase in height.

Acromegaly: Excess secretion of growth hormone in adults may lead to abnormal enlargement of head, face, hands and feet.

10th_Science Unit-16.indd 233 12/3/2022 3:04:32 PM